

v5 event-jet flavour tagging

Signal pilot, opposite-jet Z-flavour check, and saved candidate variables

FCC-ee analysis review

6 October 2026

Diagnostic studies only; no tag cut or BDT has been adopted.

Question and study scope

Physics question

Can the flavour of the event jet opposite a reconstructed $\Lambda_b \rightarrow \Lambda^0 \gamma$ candidate help distinguish signal-like $Z \rightarrow b\bar{b}$ events from selected $Z \rightarrow s\bar{s}$ and $Z \rightarrow c\bar{c}$ candidates?

- Run the existing reconstructed candidate selection and attach pretrained Weaver scores to two exclusive-ee- k_t jets.
- Associate each candidate to its nearest jet by direction; call the other jet the opposite jet.
- Keep truth as an evaluation label. No FT score is a Stage 1 selection.

Scope

The pilot is unweighted and low statistics. It tests saved variables and a separation hypothesis; it does not estimate inclusive rejection or expected yields.

Implementation: `analysis/studies/lb2lambdaga_reco_v5.py`. Study definitions: `docs/STAGE1_V5_FLAVTAG_VARIABLES.md`.

Processing and selections

- 1 Build reconstructed $p\pi$ and photon candidates with the v5 reconstruction config; include both charge conjugates.
- 2 Keep the established fitted- Λ^0 mass and vertex requirements. Apply the $E(\Lambda_b) \geq 10.5$ GeV event gate after candidate building and before truth annotation and tagger inference.
- 3 Cluster all reconstructed particles into two event jets; supply the reconstructed PV to Weaver and save class scores per jet.
- 4 Copy the nearest-jet and other-jet scores onto each candidate; flatten to one row per candidate with event and candidate keys.

Known modelling caveats

The pretrained model used MC PV inputs upstream; this study supplies a reconstructed PV. Candidate daughters are not removed before jet clustering. Both choices need calibration checks.

Config: config/lb_reco_v5_training.json. Shared offline region in reports: fitted $|m(p\pi) - 1.115683| \leq 12.5$ MeV and $E(\Lambda_b) \geq 10.5$ GeV.

Signal pilot: the variables survive normal processing

- 1,000 Physics input events; 584 candidate-bearing output events.
- 627 flattened candidate rows; 572 truth-matched signal rows.
- Shared offline region: 612 candidates, including the 572 matched rows.
- All ten B/C/S/Q/G candidate-aligned score columns are present with no nulls in that region.

Other-jet b-score retention

Threshold	Matched signal retained
≥ 0.50	91.4%
≥ 0.80	88.5%
≥ 0.90	85.3%
≥ 0.95	80.1%
≥ 0.99	40.7%

Interpretation

These are signal-only raw score efficiencies. They measure no background rejection and are not cut recommendations.

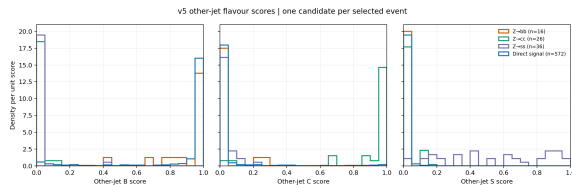
Six-file Z-flavour pilot: opposite jet is flavour-sensitive

Sample	Inputs	Rows in region	Events
Zbb	6,000	16	16
Zcc	6,000	29	26
Zss	6,000	39	36

Largest other-jet B/C/S score predicted the generated sample flavour for 74/78 selected events.

Raw other-jet $b \geq 0.8$ diagnostic

Direct signal: 506/572 (88.5%); Zbb: 13/16; Zcc: 0/26; Zss: 0/36.



Curves are unit-normalized score shapes, not event yields. Six files and first 1,000 entries per file; no cross-section weights. See docs/STAGE1_V5_ZFLAVOUR.PILOT_2026-10-06.md.

Candidate-aligned flavour-tag scores

Class X	Exact candidate branches	Meaning
B	lb_flavtag_v5_recojet_isB_flavtag_v5_associated	Candidate-side jet b score; other-jet b score.
	lb_flavtag_v5_recojet_isB_flavtag_v5_otherjet_max	
C	lb_flavtag_v5_recojet_isC_flavtag_v5_associated	Candidate-side jet c score; other-jet c score.
	lb_flavtag_v5_recojet_isC_flavtag_v5_otherjet_max	
S	lb_flavtag_v5_recojet_isS_flavtag_v5_associated	Candidate-side jet s score; other-jet s score.
	lb_flavtag_v5_recojet_isS_flavtag_v5_otherjet_max	
Q	lb_flavtag_v5_recojet_isQ_flavtag_v5_associated	Candidate-side jet light-quark score; other-jet light-quark score.
	lb_flavtag_v5_recojet_isQ_flavtag_v5_otherjet_max	
G	lb_flavtag_v5_recojet_isG_flavtag_v5_associated	Candidate-side jet gluon score; other-jet gluon score.
	lb_flavtag_v5_recojet_isG_flavtag_v5_otherjet_max	

associated means the jet nearest the reconstructed candidate direction. otherjet_max is the maximum score among the remaining jets (the opposite jet for the default two-jet setup). Missing association/score is encoded as -1 .

Per-jet vectors are saved; the proposed BDT uses six B/C/S scores.

What is stored per candidate beyond the ten FT scores?

Candidate family	Saved information
Identity and truth labels	<code>event_entry</code> , candidate slot/multiplicity, charge sign, reconstructed-to-MC match and ancestry labels. Truth is attached after candidate building.
Candidate kinematics	Four-momenta and masses for proton, pion, fitted Λ^0 , photon and Λ_b ; helicity angle and thrust/hemisphere observables.
Vertex and displacement	Fitted PV/SV quantities, vertex quality, daughter d_0 significances, Λ^0 flight and impact-parameter observables.
Event activity	Photon- and Λ^0 -centred isolation, recoil/balance and photon-partner diagnostics; these remain available independently of the FT scores.
Flavour tagging	Ten candidate-aligned B/C/S/Q/G score columns shown on the preceding slide, plus event-level jet vectors and kinematics.

Candidate granularity

The Stage 1 ROOT tuple stores candidate vectors per event. Flattening produces one row per candidate and retains event keys and candidate multiplicity.

Limits and next decision

- Only 78 selected $Z_{bb}/Z_{cc}/Z_{ss}$ events survive the pilot region; 0/36 Z_{ss} events passing the b-score cut is not evidence of zero inclusive Z_{ss} background.
- Opposite b tagging also retains 13/16 selected Z_{bb} candidates, as expected for a signal produced in $Z \rightarrow b\bar{b}$.
- No production weights were used; file IDs and entry ranges are a bounded pilot, not an inclusive mixture.
- Weaver inputs use reconstructed PV and include candidate daughters in the clustered jets; calibrate both effects.

Recommended next comparison

Train a combined BDT with the six B/C/S associated/other-jet scores plus the frozen offline feature set. Evaluate on independent, larger v5 $Z_{bb}/Z_{cc}/Z_{ss}$ samples at matched signal efficiency, keeping direct signal, wrong combinations, and each background separate.

The PI decides whether a later threshold or model becomes a selection scenario.

No v5 Condor jobs have been submitted. The Z-flavour Condor checks were check-only.